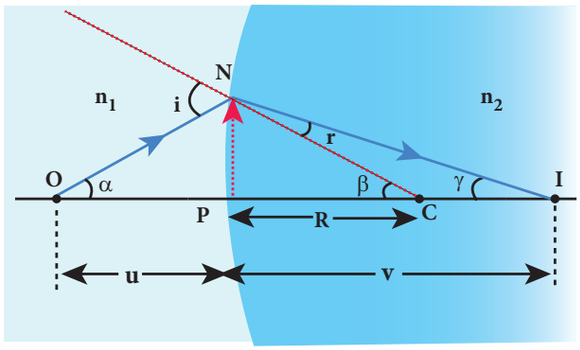


Figure 6.31 Refraction at single spherical surface

Light from O falls on the refracting surface at N. The normal drawn to the refracting surface at the point of incidence passes through the centre of curvature C. As $n_2 > n_1$, light in the denser medium deviates towards the normal and meets the principal axis at I where the image is formed.

Snell's law in product form for the refraction at the point N can be written from equation (6.19),

$$n_1 \sin i = n_2 \sin r$$

As the angles are small, sine of the angle could be approximated to the angle itself.

$$n_1 i = n_2 r \qquad (6.51)$$

Let the angles be,

$\angle NOP = \alpha$, $\angle NCP = \beta$, $\angle NIP = \gamma$

From the right angle triangles ΔNOP, ΔNCP and ΔNIP,

$$\tan\alpha = \frac{PN}{PO}; \quad \tan\beta = \frac{PN}{PC}; \quad \tan\gamma = \frac{PN}{PI}$$

As these angles are small, tan of the angle could be approximated to the angle itself.

$$\alpha = \frac{PN}{PO}; \ \beta = \frac{PN}{PC}; \ \gamma = \frac{PN}{PI} \qquad (6.52)$$

For the triangle, ΔONC,

$$i = \alpha + \beta \qquad (6.53)$$

For the triangle, ΔINC,

$$\beta = r + \gamma \text{ (or) } r = \beta - \gamma \qquad (6.54)$$

Substituting for i and r from equations (6.53) and (6.54) in the equation (6.51).

$$n_1(\alpha + \beta) = n_2(\beta - \gamma)$$

After rearranging,

$$n_1\alpha + n_2\gamma = (n_2 - n_1)\beta$$

Substituting for α, β and γ from equation (6.52),

$$n_1\left(\frac{PN}{PO}\right) + n_2\left(\frac{PN}{PI}\right) = (n_2 - n_1)\left(\frac{PN}{PC}\right)$$

Further simplifying by cancelling PN,

$$\frac{n_1}{PO} + \frac{n_2}{PI} = \frac{n_2 - n_1}{PC} \qquad (6.55)$$

PI, PO and PC are mere distances; here we use sign conventions so that the expression that is derived will be a general one. The general expression thus obtained will be valid for the situations other than the one shown in figure.

$\therefore$ substituting PO = $-u$, PI = $+v$ and PC = $+R$ in equation (6.55),

$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{(n_2 - n_1)}{R}$$

After rearranging, finally we get,

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R} \qquad (6.56)$$

Equation (6.56) gives the relation among the object distance u, image distance v, refractive indices of the two media (n_1 and n_2) and the radius of curvature R of the spherical surface. It holds good for any spherical surface as sign conventions are applied.

If the first medium is air, then $n_1 = 1$ and for the second medium $n_2 = n$, then the equation is reduced to,

$$\frac{n}{v} - \frac{1}{u} = \frac{(n-1)}{R} \qquad (6.57)$$

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